

Installing Beliefs to Audit Attribution: A Ground-Truth Testbed and Five Ways It Misleads

Anonymous

Abstract

Across three-seed bands under the same premise specification as the long-form arms, with form as the only variable, the short-dense belief contrast was 0.119. In a separate one-seed, long-sparse evidence-only condition with a matched off-topic control and all arms gated, the contrast rose from 0.0072 at optimizer step 24 after two epochs—the near-null cell quoted in earlier drafts—to 0.0342 at step 36 after three epochs with the same arms and control. At the later checkpoint, all seven arms gated and the control passed at 0.802, making the gate failure genuinely late.

Control behavior varied by adaptation regime. For 4B LoRA, the machinery quantity subtracted by netting changed from -0.0173 at seed 42 to -0.0025 at seed 7, with nonoverlapping seed intervals. Under full fine-tuning, it was -0.0053 at seed 42 and -0.0049 at seed 7, with overlapping intervals unlike every LoRA control family. For the second topic, software architecture, netted belief contrasts were 0.0076 at seed 42 with a matched off-topic control and all arms gated, 0.0122 in the seed-7 replicate with a seed-matched control, and -0.0034 at seed 123 with a seed-matched control and all five arms gated. The seed-123 sign disagreed with the other two because of the control machinery term rather than the treatment contrast.

The descriptive-inference explicit-arm positive-control contrast was 0.062, a point estimate derived from recorded arm scores without an inferred interval. Because the arm intended to be most sensitive could not demonstrate premise-specificity, this was an instrument-failure result: the descriptive-inference axis was withdrawn and no transfer number from it was reported. The per-facet ranking was computed outside the declared evidence, so no ranking was asserted.

1 Motivation

Absorption is held-out premise specialization, not a belief measure.

2 Methodology

The analysis was exploratory, with model, topic, seed count, and checkpoint scope restricted to recorded artifacts. Two-seed scope was used only for declared seed-paired contrast families, while single-seed and instrument-specific readings were identified separately.

Belief and action contrasts were netted against their matched off-topic controls and were not interpreted as evidence of causal mediation. Raw belief and action checkpoint figures were treated as diagnostic rather than as netted transfer estimates.

Fixed tables were treated as authoritative for numerical estimates and confidence intervals rather than the prose.

3 Results

In the single-seed long-sparse evidence-only condition with matched off-topic control and all arms gated, the belief contrast rose from 0.0072 at optimizer step 24 after two epochs to 0.0342 at step 36 after three epochs. The earlier reading was the near-null cell quoted in earlier drafts. At step 36, all seven arms gated and the control passed at 0.802, indicating that the gate failure was genuinely late. By contrast, the explicit-stance positive-control belief contrast, which used a lower token dose than the evidence corpus and indexed the

belief instrument rather than general belief transfer, declined from 0.3111 at step 24 to 0.2167 one epoch later while the evidence cell was still rising; these positive-control estimates were derived from recorded arm scores without inferred intervals.

In the form cross, the long-dense belief contrast was 0.0547 in the length arm at matched density, whereas the short-dense contrast was 0.119 across three-seed bands under the same premise specification, with form as the only variable. The short-sparse contrast was 0.0506, but this unstable corner showed 2.98x scatter across three seeds, making the result directional and precluding a ratio.

For dose-matched explicit-stance conditions, with model as the only difference, the recorded positive-control belief contrast was 0.1517 at 4B. At 8B, with a retrained control and its own calibrated gate threshold rather than the 4B threshold, the belief contrast was 0.097. The corresponding action contrast was 0.0011 at 4B and its interval straddled zero, so no conduction ratio was quotable. At 8B, the action contrast was 0.0352 and its interval excluded zero; its direction replicated at seed 7, but 44 percent cross-seed scatter precluded a magnitude claim.

For the evidence corpus under full fine-tuning, the belief contrast was 0.0164 at seed 42 and 0.0115 in the seed-7 replication, which used a seed-matched full-fine-tuning control and re-derived schedule. Under LoRA, the same corpus straddled zero on the probability scale but excluded zero on log-odds, making the contrast scale-dependent rather than evidence of a capacity cliff. The netting machinery differed more across seeds under LoRA: at 4B it changed from -0.0173 at seed 42 to -0.0025 at seed 7, with nonoverlapping intervals, and at 8B it changed sign from -0.0031 to 0.0156, again with nonoverlapping intervals. Under full fine-tuning, the machinery term was -0.0053 at seed 42 and -0.0049 at seed 7, and the intervals overlapped, unlike every LoRA control family.

For the second topic, software architecture, the netted belief contrast was 0.0076 at seed 42 with a matched off-topic control and all arms gated, 0.0122 in the seed-7 replicate with a seed-matched control, and -0.0034 at seed 123, where all five arms gated. The seed-123 sign disagreed with the other two seeds; this disagreement arose in the control machinery term rather than the treatment contrast.

The descriptive-inference explicit-arm positive-control contrast was 0.062, a point estimate derived from recorded arm scores without an inferred interval. Because the arm intended to be most sensitive could not demonstrate premise specificity, this was an instrument-failure result: the descriptive-inference axis was withdrawn and no transfer estimate from it was reported. The per-facet ranking was computed outside the declared evidence, so no ranking was asserted.

Table 1: Run ms3p_arms suite summary; identity rendering; short-dense focal estimate and interval with form-cross interpretation linked to separate declared runs in the ladder. Source: ms3p_arms/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.1034 [+0.0741, +0.1352]
machinery	-0.0156 [-0.0271, -0.0048]
delta_net	+0.1190 [+0.0901, +0.1497]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.1825

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 2: matrix_v1_step24 belief suite; optimizer step 24; identity values; no interval inferred for the derived positive-control contrast. Source: matrix_v1_step24/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0033 [-0.0009, +0.0082]
machinery	-0.0039 [-0.0093, +0.0007]
delta_net	+0.0072 [+0.0016, +0.0136]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0110

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 3: matrix_v1_step36 belief suite; optimizer step 36; identity values; no interval inferred for the derived positive-control contrast. Source: matrix_v1_step36/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0228 [+0.0154, +0.0307]
machinery	-0.0114 [-0.0175, -0.0050]
delta_net	+0.0342 [+0.0248, +0.0443]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0524

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 4: h8_4b belief suite only; identity rendering; dose-matched model comparison interpreted through the ladder. Source: h8_4b/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.1344 [+0.0930, +0.1779]
machinery	-0.0173 [-0.0250, -0.0106]
delta_net	+0.1517 [+0.1109, +0.1942]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 5: h8_8b belief suite only; identity rendering; 8B-specific gate threshold retained. Source: h8_8b/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0938 [+0.0787, +0.1091]
machinery	-0.0031 [-0.0069, +0.0008]
delta_net	+0.0970 [+0.0826, +0.1115]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 6: h8_4b action suite only; identity rendering; matched off-topic netting; no causal-mediation claim. Source: h8_4b/action_summary.yaml.

quantity	recorded value
delta_raw	+0.0013 [-0.0083, +0.0100]
machinery	+0.0001 [-0.0049, +0.0049]
delta_net	+0.0011 [-0.0101, +0.0121]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 7: h8_8b action suite only; identity rendering; directional seed replication and magnitude restriction retained. Source: h8_8b/action_summary.yaml.

quantity	recorded value
delta_raw	+0.0201 [+0.0128, +0.0280]
machinery	-0.0151 [-0.0215, -0.0090]
delta_net	+0.0352 [+0.0249, +0.0459]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

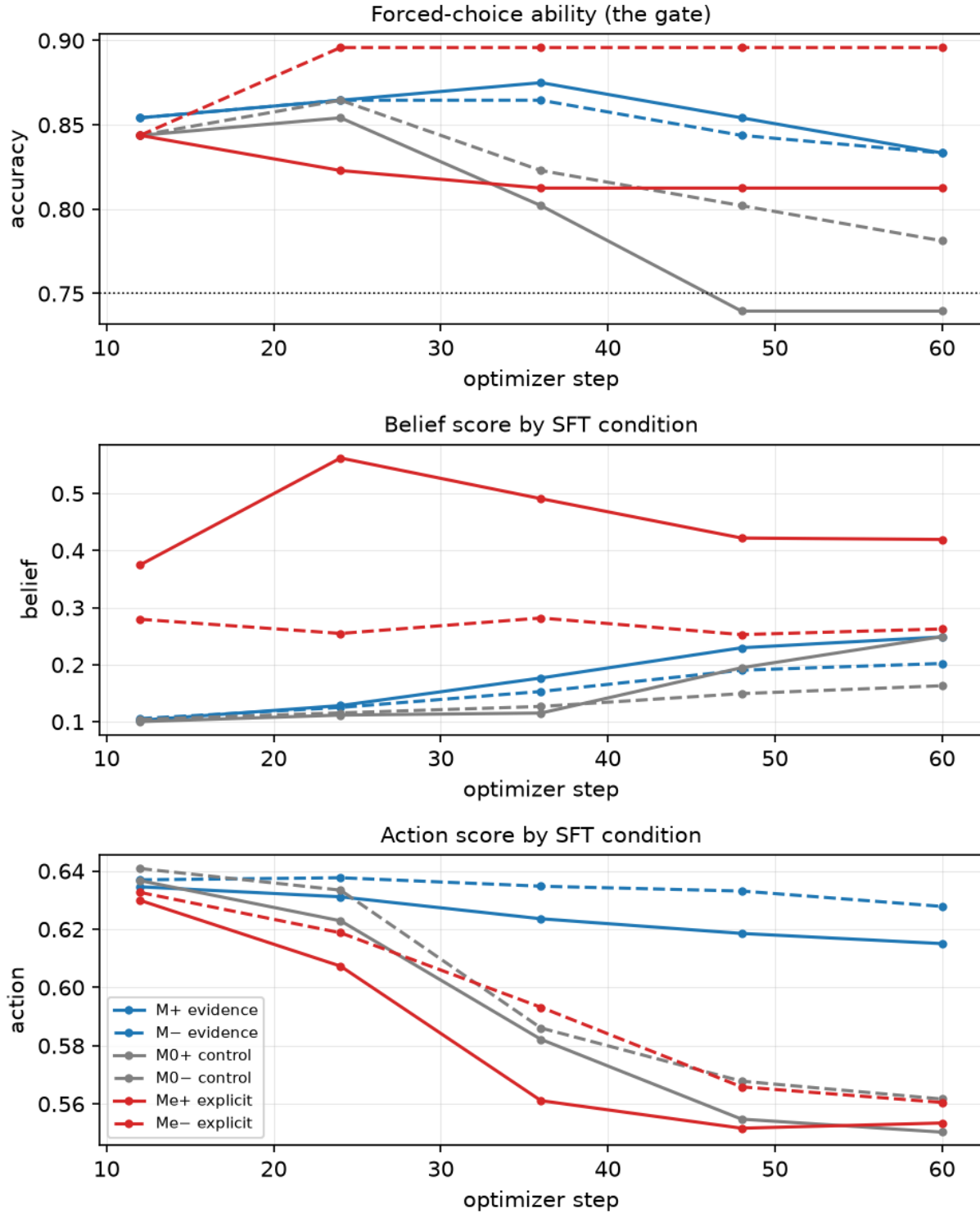


Figure 1: Saved matrix_v1 trajectory figures; diagnostic only; mark steps 24 and 36; direct readers to fixed tables for authoritative netted estimates and intervals..

4 Discussion

The full-fine-tuning evidence-corpus belief contrast was 0.0164 at seed 42 and 0.0115 in the seed-seven replication with a seed-matched full-FT control and a re-derived schedule. Under LoRA, the same corpus straddles zero on probability and excludes zero on log-odds, so this comparison is scale-dependent rather than a capacity cliff. The control-machinery term, which netting subtracts, changed from -0.0173 at seed 42 to -0.0025 at seed 7 for 4B LoRA, with nonoverlapping seed intervals. For 8B LoRA, it changed sign from -0.0031 at seed 42 to 0.0156 at seed 7; the intervals do not overlap and the readings have opposite signs. Under full fine-tuning, it was -0.0053 at seed 42 and -0.0049 at seed 7, with overlapping intervals unlike every LoRA control family.

For the second topic, software architecture, the netted belief contrast was 0.0076 at seed 42 with a matched off-topic control and all arms gated, 0.0122 at seed 7 with a seed-matched control, and -0.0034 at seed 123 with a seed-matched control and all five arms gated. These three netted signs disagree: the seed-123 sign differs from the other two because of the control machinery term rather than the treatment contrast. Two-seed scope applies only to declared seed-paired contrast families; single-seed and instrument-specific readings remain separate.

The descriptive-inference explicit-arm positive-control contrast was 0.062, a point estimate derived from recorded arm scores without an inferred interval. Because the arm intended to be most sensitive could not demonstrate premise-specificity, this is an instrument-failure result; the descriptive-inference axis is withdrawn and no transfer number from it is reported. The per-facet ranking is computed outside the declared evidence, so no ranking is asserted.

A Supplementary diagnostics

Table 9: h19_full_ft belief suite only; identity rendering; seed 42; retain scale-dependence qualification. Source: h19_full_ft/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0111 [+0.0030, +0.0212]
machinery	-0.0053 [-0.0101, -0.0014]
delta_net	+0.0164 [+0.0060, +0.0282]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0251

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 10: h19_full_ft_s7 belief suite only; identity rendering; seed-matched control and re-derived schedule. Source: h19_full_ft_s7/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0066 [+0.0012, +0.0137]
machinery	-0.0049 [-0.0090, -0.0012]
delta_net	+0.0115 [+0.0046, +0.0199]
sensitivity	+0.6521 [+0.5578, +0.7413]
T (delta_net)	+0.0176

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 8: Exactly one all-contrast ladder; declared-contrast rendering; preserve intervals, absent-interval labels, seed-family scope, checkpoints, and all qualifications. Source: paper/synthesis.json.

intervention	net effect	qualification
Second topic, evidence-only, belief, seed 42	+0.0076 [-0.0008, +0.0159]	Second topic (software architecture), matched off-topic control, all arms gated. Reported as one of three seeds whose netted signs disagree.
Second topic, evidence-only, belief, seed 7	+0.0122 [+0.0053, +0.0202]	Seed-seven replicate with a seed-matched control.
Second topic, evidence-only, belief, seed 123	-0.0034 [-0.0156, +0.0089]	Third seed with a seed-matched control, all five arms gated. Its sign disagrees with the other two seeds; the disagreement arises in the control's machinery term, not the treatment contrast.
Control machinery, 4B LoRA, seed 42	-0.0173 [-0.0250, -0.0106]	The quantity netting subtracts. Compare against the seed-7 reading of the same family.
Control machinery, 4B LoRA, seed 7	-0.0025 [-0.0078, +0.0036]	Its interval does not overlap the seed-42 reading of the same family.
Control machinery, 8B LoRA, seed 42	-0.0031 [-0.0069, +0.0008]	Compare against the seed-7 reading, whose sign differs.
Control machinery, 8B LoRA, seed 7	+0.0156 [+0.0086, +0.0226]	Its interval does not overlap the seed-42 reading, and the two carry opposite signs.
Control machinery, full fine-tuning, seed 42	-0.0053 [-0.0101, -0.0014]	The full fine-tuning control, whose two seed intervals overlap.
Control machinery, full fine-tuning, seed 7	-0.0049 [-0.0090, -0.0012]	Overlaps its seed-42 counterpart, unlike every LoRA control family.
Evidence-only, belief, step 24 (2 epochs)	+0.0072 [+0.0016, +0.0136]	The reading the earlier drafts quoted as the near-null cell. Matched off-topic control, all arms gated.
Evidence-only, belief, step 36 (3 epochs)	+0.0342 [+0.0248, +0.0443]	Same arms, same control, three epochs. All seven arms gated; the control passes at 0.802, so the late-step gate failure is genuinely late. One seed.
Explicit stance, belief, step 24	+0.3111	Positive control for the belief instrument. Token dose is lower than the evidence corpus. Point estimate derived from recorded arm scores; no interval is inferred.
Explicit stance, belief, step 36	+0.2167	The same positive control one epoch later, where it has begun to decay while the evidence cell is still rising. Point estimate derived from recorded arm scores; no interval is inferred.
Short, dense (Ms3p), belief	+0.1190 [+0.0901, +0.1497]	Same premise specification as the long-form arms; form is the only variable. Band-level across three seeds.
Long, dense (Mld), belief	+0.0547 [+0.0391, +0.0705]	The length arm of the form cross at matched density.
Short, sparse (Mss), belief	+0.0506 [+0.0308, +0.0720]	The unstable corner: scatters 2.98x across three seeds, so its reading is directional and carries no ratio.
Explicit stance at 4B, belief	+0.1517 [+0.1109, +0.1942]	Dose-matched to the 8B arm; model is the only difference.
Explicit stance at 8B, belief	+0.0970 [+0.0826, +0.1115]	Control retrained at 8B. Qwen3-8B has its own calibrated gate threshold; the 4B bar is not reused.
Explicit stance at 4B, action	+0.0011 [-0.0101, +0.0121]	Straddles zero. No conduction ratio is quotable while this numerator straddles.
Explicit stance at 8B, action	+0.0352 [+0.0249, +0.0459]	Excludes zero. Replicated in direction at seed 7; scatters 44 percent across seeds, so no magnitude.
Full fine-tuning, evidence corpus, belief, seed 42	+0.0164 [+0.0060, +0.0282]	Under LoRA the same corpus straddles zero on probability and excludes zero on log-odds, so the contrast is scale-dependent rather than a capacity cliff.
Full fine-tuning, evidence corpus, belief, seed 7	+0.0115 [+0.0046, +0.0199]	Seed-seven replication with a seed-matched full-FT control and a re-derived schedule.
Descriptive-inference suite, explicit arm (the INVERTED positive control)	+0.0620	This entry is the instrument-failure result: on the arm meant to be most sensitive the suite could not demonstrate premise-specificity, so the descriptive-inference axis is withdrawn and no transfer number from it is reported. The per-facet ranking behind that judgement is computed in scripts/h25_positive_control.py and sits deliberately outside this declared bundle. Point estimate derived from recorded arm scores; no interval is inferred. Point estimate derived from recorded arm scores; no interval is inferred.

Note. Rows without intervals are deterministic point contrasts over recorded arm scores.

Table 11: software_architecture/sw_arms_v1 belief suite only; identity rendering; matched control; all arms gated. Source: software_architecture/sw_arms_v1/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0155 [+0.0084, +0.0233]
machinery	+0.0079 [+0.0031, +0.0133]
delta_net	+0.0076 [-0.0008, +0.0159]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	+0.0113

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 12: software_architecture/sw_arms_v1_s7 belief suite only; identity rendering; seed-matched control. Source: software_architecture/sw_arms_v1_s7/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0217 [+0.0115, +0.0333]
machinery	+0.0094 [+0.0038, +0.0152]
delta_net	+0.0122 [+0.0053, +0.0202]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	+0.0183

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 13: software_architecture/sw_arms_v1_s123 belief suite only; identity rendering; all five arms gated; pre-serve control-machinery explanation. Source: software_architecture/sw_arms_v1_s123/belief_summary.yaml.

quantity	recorded value
delta_raw	+0.0148 [+0.0050, +0.0264]
machinery	+0.0182 [+0.0094, +0.0280]
delta_net	-0.0034 [-0.0156, +0.0089]
sensitivity	+0.6691 [+0.5550, +0.7763]
T (delta_net)	-0.0050

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.

Table 14: inference_v1_step24 inference suite only; identity arm scores; derived point estimate labeled unintervalled; omit repository-only per-facet ranking. Source: inference_v1_step24/inference_summary.yaml.

quantity	recorded value
delta_raw	+0.0134 [+0.0043, +0.0245]
machinery	+0.0013 [-0.0049, +0.0075]
delta_net	+0.0121 [+0.0012, +0.0245]

Note. Intervals are paired bootstrap CIs from the stored summary; bold excludes zero.